

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Principles of pollution control (CPA, CGM, new technologies) - Set 1

Mark scheme

AQA spec 3.4.3.1, 3.4.3.3; Pollution; textbook pp. 243-245, 303-304

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) (1) • To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites (1)	2
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sea current velocity and direction (1) • Properties of the radionuclides: half-life, solubility, tendency to adsorb onto sediments / bioaccumulate (1) • Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people (1) • Geology / hydrology of the coast; sediment deposition areas (mud flats) (1) • Human activities in the area: fishing, seaweed collection, beach use (1)	3
01.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) (1) • Monitor their exposure / the concentration of radioactive material in the foods they eat or their body (1) • If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk; if not, emissions are controlled (1) • CGM assesses exposure before health effects occur (not used to detect damage) (1)	3

Question	Marking guidance	Marks
01.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Seawater (1) • Sediment / mud (1) • Seaweed (1) • Shellfish / molluscs / fish (1) • Milk, vegetables or meat from nearby farms (1) • Atmospheric dust (1) • Soil (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
02.1	<i>1 mark for each point.</i> • Location: discharge into the sea / large water bodies where currents dilute and disperse the pollutant (1) • Location: chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations; avoiding valleys and enclosed water bodies where pollutants concentrate (1) • Location: not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers (1) • Timing: restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor (1) • Timing: discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away (1) • Timing: pesticide spraying at night / on still days; noisy operations restricted to daytime; night-flight restrictions (1)	6
02.2	<i>1 mark for each point.</i> • The organisation / person that causes the pollution is responsible for (paying for) the damage caused (1) • If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it; works best when penalties are high and escape is unlikely (1)	2
02.3	<i>1 mark for each point.</i> • It is assumed that the waste will cause pollution / harm until research shows it is unlikely to (1) • So release is not permitted until it has been tested; lack of knowledge of a problem is not an excuse / does not reduce responsibility (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
03.1	<i>1 mark for each point.</i> • Cost increases as the percentage prevented increases (1) • The relationship is not linear: cost rises slowly at first then very steeply / exponentially above about 80-90% (1)	2
03.2	<i>1 mark for the correct answer.</i> • $16.0 - 5.0 = 11.0$ (£ million) (1)	1

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> • ALARA: emissions should be As Low As Reasonably Achievable (1) • It is rarely practical / affordable to reduce emissions to zero; low-level emissions may have acceptable impacts so an acceptable emission level is set (1) • BATNEEC: Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive (1) • The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% (1)	4
03.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Industry / city may have expanded so there are more pollutant sources and total emissions have risen (1) • Better understanding of the health / environmental effects so stricter standards are set (1) • Improved technology becomes available (BATNEEC changes) / cost of better control falls (1) • Pollutant may have accumulated (persistent) so background levels are higher (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
04.1	<i>1 mark for each point.</i> • Production prevention: desulfurisation of fuels before combustion; low-temperature (fluidised bed) combustion to prevent NO_x; smokeless fuels (1) • Prevention of release: electrostatic precipitators / bag filters / cyclones for smoke; catalytic converters; flue gas desulfurisation; bund walls; landfill liners (1) • Post-release remediation: oil spill clean-up (booms, skimmers, dispersants, bioremediation); phytoremediation of metal-contaminated land; liming of acidified lakes (1) • Alternative processes: electric vehicles instead of petrol / diesel; renewable energy instead of fossil fuels; pyrethroids instead of organochlorines; fuel cells instead of internal combustion engines (1)	4
04.2	<i>1 mark for each point.</i> • Remediation is often incomplete / cannot recover all the pollutant (eg oil spills, dispersed pollutants), so some damage still occurs (1) • Prevention avoids the damage entirely and is usually cheaper than clean-up / avoids polluter-pays costs; moving from post-production treatment to non-release is a circular economy principle (1)	2
04.3	<i>1 mark for each point.</i> • Satellites detect and track the extent / movement of oil slicks over large areas, quickly and repeatedly, so clean-up can be directed and polluters identified (1) • GPS tracking monitors ship positions so navigation errors / collisions / groundings are reduced and offshore shipping routes are enforced (1) • Polymers with a high surface area adsorb metal ions from contaminated water / effluent so they can be removed and recovered (1) • Plants that absorb heavy metals (hyperaccumulators) are grown on contaminated land, harvested and disposed of / metals recovered, cleaning the soil (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
05.1	<i>Correct answer only.</i> • C (1)	1
05.2	<i>1 mark for each point.</i> • Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate (1) • No need to sample sites that were never likely to be polluted; sampling also checks the accuracy of the predictions (1)	2
05.3	<i>1 mark for each point.</i> • Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify (1) • So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
06.1	<i>Correct answer only.</i> • A waste is assumed to cause pollution until research shows it is unlikely to do so (1)	1
06.2	<i>1 mark for each point.</i> • If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution (1) • If polluters can escape their responsibilities / penalties are low, there is little incentive to comply (1)	2
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marine discharges where currents dilute and disperse emissions (1) • Emissions downwind of urban areas (1) • Not discharging waste onto permeable rock above an aquifer (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
07.1	<i>1 mark for each point.</i> • Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) (1) • It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people (1)	2
07.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Physical state of the effluent (solution / solid) (1) • Ocean currents / river flow (1) • Effect of pH and oxygen on solubility (1) • Bioaccumulation and food-chain concentration (1) • Food sources and consumption by local people (eg shellfish, seaweed) (1) • Half-lives of the isotopes (1) • Meteorological conditions for any gaseous release (1) • Nature of the environment - geology, vegetation (1)	4
07.3	<i>1 mark for each point.</i> • Monitoring focuses on the sites most at risk of contamination (1) • No need to sample sites that were never likely to be polluted; sampling checks the accuracy of the predictions (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> - It assesses the risk to the people most exposed because of their lifestyle before any health effects occur (1) - If their risk is acceptably low everyone else has a lower risk; if unacceptable, emissions are reduced to prevent future exposure (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
08.1	<i>1 mark for the correct answer.</i> $16.0 - 5.0 = \text{£}11 \text{ million}$ (1)	1
08.2	<i>1 mark for each point.</i> - It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided (1) - ALARA: emissions should be As Low As Reasonably Achievable (1) - BATNEEC: the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this (1)	3
08.3	<i>1 mark for each point.</i> - As industries expand or cities grow the number of pollutant sources increases (1) - So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed (1)	2

Question	Marking guidance	Marks
08.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Critical Pathway Analysis: predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk; mainly radioactive discharges; checked by sampling • Critical Group Monitoring: exposure of the most-exposed group monitored before health effects; if their risk is acceptable everyone else is safer; emissions reduced if not • Emission control strategies: location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) • Principles: polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship • Selection of methods: production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) • Evaluation: prevention usually cheaper and more effective than remediation; costs rise steeply near 100%; CPA relies on understanding of pathways; enforcement and international agreement needed for transboundary pollutants; judgement about acceptable risk is political as well as scientific • Justified conclusion	9
	Model answer - Level 3 (8 marks)	
	Deciding how much to control pollution starts with predicting where a pollutant will go. Critical Pathway Analysis uses the state of the effluent, wind and currents, geology, the effect of pH and oxygen on solubility, bioaccumulation and local food sources to predict whether a discharge will disperse harmlessly or concentrate somewhere sensitive; it is mainly used for radioactive discharges and means monitoring can focus on the sites at risk rather than sampling everywhere. Critical Group Monitoring then checks the exposure of the people most at risk from their lifestyle before any health effects occur: if their risk is acceptable everyone else's is lower, and if not, emissions are reduced.	

Question	Marking guidance	Marks
	The choice of control is guided by principles. The polluter pays principle gives a financial incentive to prevent damage, but only if penalties are high and polluters cannot escape responsibility. The precautionary principle assumes a waste is harmful until research shows otherwise, so ignorance is no defence. Because the cost of control rises steeply as emissions approach zero, emissions are set As Low As Reasonably Achievable using the Best Available Technology Not Entailing Excessive Cost. Where emission location and timing can be chosen (marine outfalls, downwind of towns, not over aquifers; avoiding temperature inversions) severity is reduced cheaply.	
	Overall, prevention of production (fuel desulfurisation) or of release (electrostatic precipitators, catalytic converters) is more effective than post-release remediation such as oil-spill clean-up, which never recovers all the pollutant. CPA depends on understanding pathways, and transboundary pollutants need international enforcement, but together these principles give a rational basis for deciding acceptable risk rather than aiming for an unaffordable zero.	
	Weaker answer - Level 1 (3 marks)	
	Pollution can be controlled in lots of ways. Critical Pathway Analysis looks at where pollution goes and Critical Group Monitoring looks at people. The polluter pays principle means the person who pollutes has to pay. The precautionary principle means being careful. Pollution can be prevented, or cleaned up afterwards with things like booms for oil spills. Electric cars are an alternative process. Pollution control can be expensive so ALARA is used. All of these are good ways of controlling pollution.	
	What makes the difference	
	• The Level 3 answer explains <i>how</i> each principle works (why penalties matter, why costs rise near zero emissions); the weak answer only names them.	
	• The Level 3 answer groups the material logically (predict, decide, select) rather than listing everything at the same weight.	
	• 'Evaluate' needs comparison and a judgement: prevention vs remediation, CPA's dependence on understanding pathways, and a conclusion about acceptable risk. The weak answer ends with 'all of these are good'.	
	• Precise examples (desulfurisation, precipitators, catalytic converters, temperature inversions) show applied knowledge; 'things like booms' does not.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	